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***B.Tech. (Part-Time) Degree III Semester Examination
April 2019***

**ME 17P 1304 MACHINE TOOLS AND MACHINING SCIENCE
(2017 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Write any four operations that can be performed using lathe machine.
- (b) How do you get reciprocating motion in a slotting machine?
- (c) List the differences between drilling and boring operation.
- (d) Explain the difference between up-milling and down-milling.
- (e) Explain (i) Lapping (ii) honing.
- (f) What are the basic rules for applying clamping forces in a fixture?
- (g) What are the variables affecting the tool life?
- (h) Define (i) machinability (ii) machinability index.
- (i) What are the properties of an ideal cutting tool material?
- (j) Write down the Lee & Shaffer's relationship in the theory of metal cutting.



PART B

(4 × 10 = 40)

- II. Explain the differences between Capstan lathe and Turret lathe with the help of neat sketches. (10)

OR

- III. Explain the crank and slotted link quick return mechanism with a neat sketch. (10)

- IV. (a) Describe a grinding wheel structure, with the help of a neat sketch. (5)
- (b) Briefly explain differential indexing in a milling machine. (5)

OR

- V. (a) What are the different types of abrasives used in grinding? How are they selected? (4)
- (b) In a steel sheet of 30 mm thickness, 4 holes of 20 mm diameter are to be drilled. The cutting speed is 32 m/min and the feed rate 0.12 mm/revolution. Calculate the machining time and the material removal rate. (6)

- VI. (a) Determine the values of normal rake of the turning tool whose geometry is designated as: 10° , -10° , 8° , 6° , 15° , 30° , 0 (inch). (5)
- (b) Explain the thermal aspects of machining. (5)

OR

- VII. (a) Explain the mechanism of built-up edge formation. (5)
- (b) In an orthogonal machining operation, the rake angle of the tool was 15° , and shear angle 35° . Calculate shear strain. (5)

(P.T.O)

- VIII. For turning a given steel rod by a tool of given geometry if shear force P_s , frictional force F and shear angle γ^0 could be estimated to be 400 N and 300 N respectively, then what would be the possible values of p_x , p_y and p_t ? (10)
(Use Merchant's Circle Diagram)

OR

- IX. (a) Two single point lathe tools made of HSS and carbide have the same tool life of 60 minutes at a cutting speed of 70 m/min. The exponent of tool life in Taylor's equation (n) is 0.12 for HSS while it is 0.21 for carbide. Compare the life of the two tools at a speed of 85 m/min. (7)
- (b) Comment about the situations where diamond is used as a cutting tool. (3)
